

Required Libraries: Install & Import

Credit Card Approval Prediction System | Epic 2: Importing Required Libraries

One-Step Install

```
pip install -r requirements.txt
```

requirements.txt (pinned versions):

```
Flask==3.1.3          gunicorn==26.0.0      joblib==1.5.3
numpy==2.4.1          pandas==2.3.3         scikit-learn==1.8.0
scipy==1.17.0         threadpoolctl==3.6.0  Werkzeug==3.1.8
Jinja2==3.1.6         MarkupSafe==3.0.3     click==8.3.1
itsdangerous==2.2.0   blinker==1.9.0
```

Library Reference

Library	Version	Used In	Purpose
Flask	3.1.3	app.py	Web framework: routing, form parsing, template rendering
joblib	1.5.3	app.py, train_model.py	Serialize/deserialize the full sklearn Pipeline (best_credit_card_model.pkl)
pandas	2.3.3	app.py, train_model.py	Load CSVs, merge datasets, build input DataFrame for prediction
numpy	2.4.1	train_model.py	Numeric array operations during training
scikit-learn	1.8.0	train_model.py	Pipeline, ColumnTransformer, StandardScaler, OneHotEncoder, RandomForestClassifier
scipy	1.17.0	train_model.py	Dependency of scikit-learn sparse matrix ops
gunicorn	26.0.0	Procfile	Production WSGI server for Render deployment
Werkzeug	3.1.8	Flask (dep)	WSGI utilities, request/response handling
Jinja2	3.1.6	Flask (dep)	HTML template engine for index.html / result.html

Imports in app.py

```
from flask import Flask, render_template, request
import pandas as pd
import joblib
```

Imports in train_model.py

```
import pandas as pd
import joblib
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier # optional
```

```
from sklearn.metrics import (accuracy_score, precision_score, recall_score,  
                             f1_score, roc_auc_score, confusion_matrix, classification_report)
```

Verify Installation

```
python -c "import flask, pandas, joblib, sklearn, gunicorn; print('All libraries OK!')"
```